

Real-Time AI & Software Pipeline for Smart Extubation Early Warning

25-channel capacitive matrix ingestion, spatial-temporal classification and clinical annunciation

- Warns before an unplanned extubation, instead of recording it afterwards.
- Browser bridge, server-side inference, and an annunciator that decides when to speak.
- 81 bench recordings: every pull episode caught, at 7.8 alarms an hour.

Asynchronous server pipeline

Gradient-boosted tree ensemble

Browser serial ingest

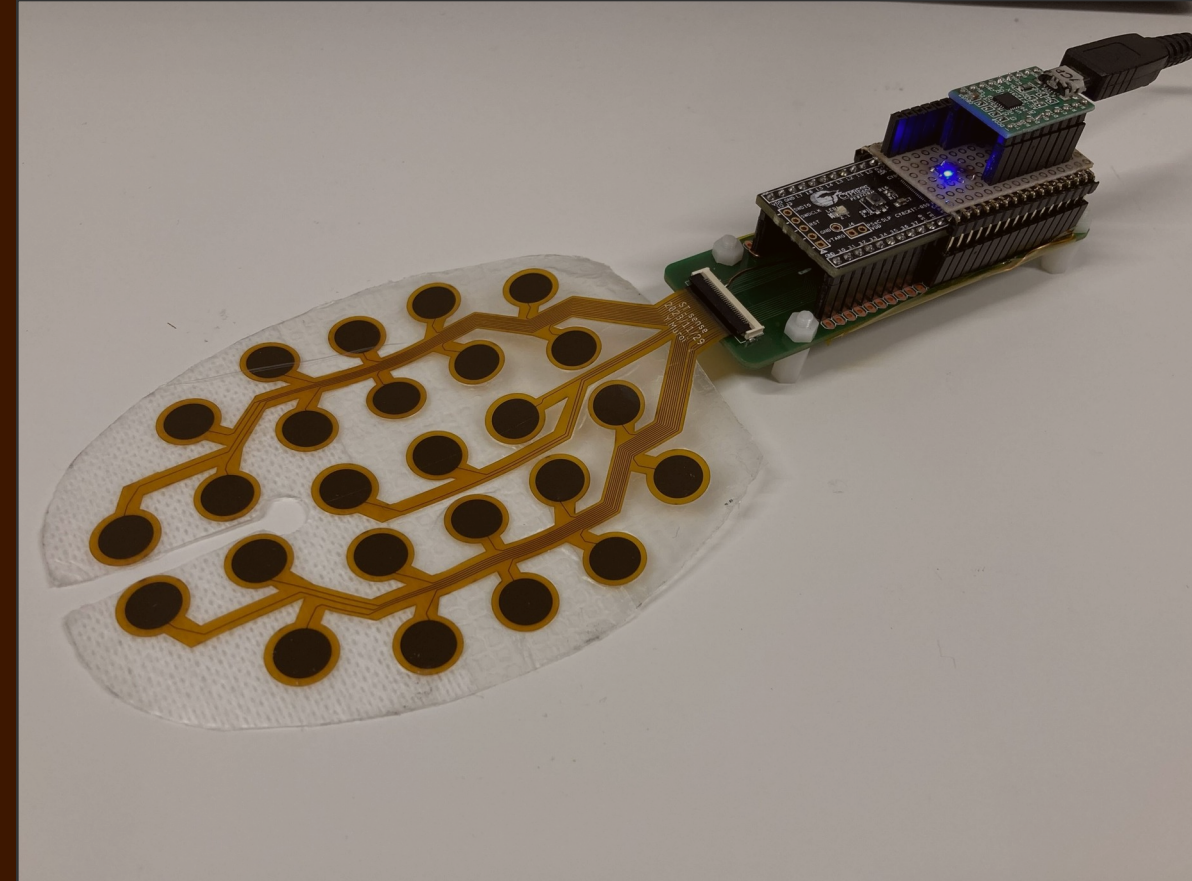
0.127 ms per frame

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The 25-pad dressing and its microcontroller board, from our own bench.



Background (1/2)

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What an unplanned extubation costs

What it is

A patient removes their own breathing tube.

What follows

Bleeding and infection. Re-intervention, and the cost that comes with it.

Why it cannot wait

With an endotracheal tube, removal can be serious immediately.

Background (2/2)

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Every countermeasure in use today trades one harm for another

Nursing rounds

High burden on staff

Reinforced fixation

Skin inflammation

Physical restraint

Ethical problems

There are limits to what the ward can do on its own.

Automatic detection is the remaining route.

Previous Research

4

Three attempts, and what each one leaves open

RFID tag on the line

Nakajima & Takahashi, 2015

Detects removal, not the approach to it.

Camera and deep learning

Yang et al.

Early warning shown, at the cost of continuous video of a patient.

Our own capacitive dressing

KES 2025

Autoencoder anomaly detection. F1 0.882 vertical, 0.833 horizontal, with false detections.

The sensor has never been characterised as a sensor. Every result so far is an algorithm result on an uncharacterised front end.

Purpose

5

What this half of the work sets out to do

01

Turn 25 numbers into one clinical decision

A pipeline that reads the dressing continuously, judges every frame, and decides when the bedside should hear about it.

02

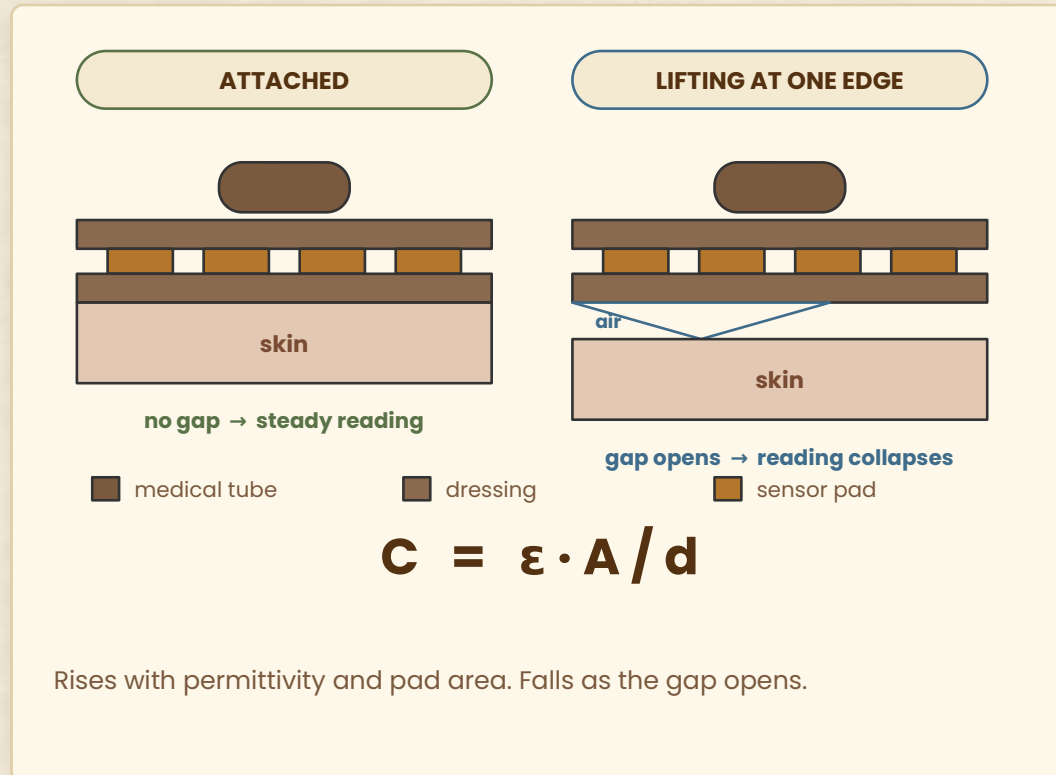
Report what the signal actually supports

Including a negative result. Where the sensor cannot see, the deck says so rather than reporting an accuracy around it.

Sensor Structure

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What the pad measures, and the physics the algorithm is anchored in



No public benchmark exists

No open dataset exists. Every choice is justified by measured physics on this patch.

Dielectric loading — touch

Tissue permittivity $\approx 50-80$ against ≈ 1 for air. Presses measure +456 to +1,717 counts.

Air gap opens — peel

Air under the pad, and the reading collapses. Median -866 counts, 10 of 10 cross -300.

Shear without lift-off — horizontal pull

Tension travels, but no gap opens. Median -64 counts, 0 of 10 cross -300.

59.85 counts per picofarad — so -300 counts is about -5 pF, and the +60 gate about +1 pF.

Proposed Method (1/7)

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Five physical interaction scenarios, and the one that refuses to separate

1

Baseline

Patient at rest

Thermal drift only,
within 60 counts

Safe · Level 0

2

Touching

Accidental finger tap

Sharp local peak,
one or two pads

Advisory · Level 1

3

Rubbing

Blanket or gown friction

Travelling multi-pad wave,
fast recovery

Advisory · Level 1

4

Pulling

Grasping and pulling the tube

Sustained multi-pad load,
high signal energy

Critical · Level 3

5

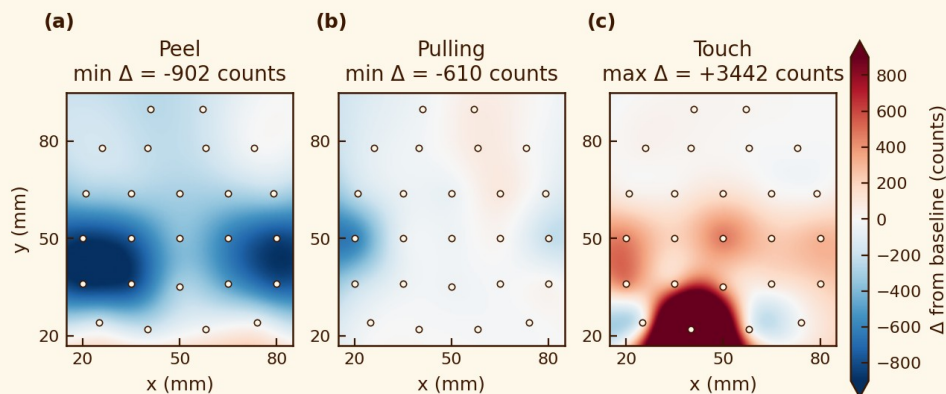
Peeling

Edge detachment

Deep negative trough,
below -300 counts

Warning · Level 2

Three of them, measured on our own recordings



The scenario that does not separate

Median deepest -64 counts — quieter than Friction, which is a normal class. 0 of 10 cross -300.

What moves is the contact side, +456. So it reads as a press, not a detachment.

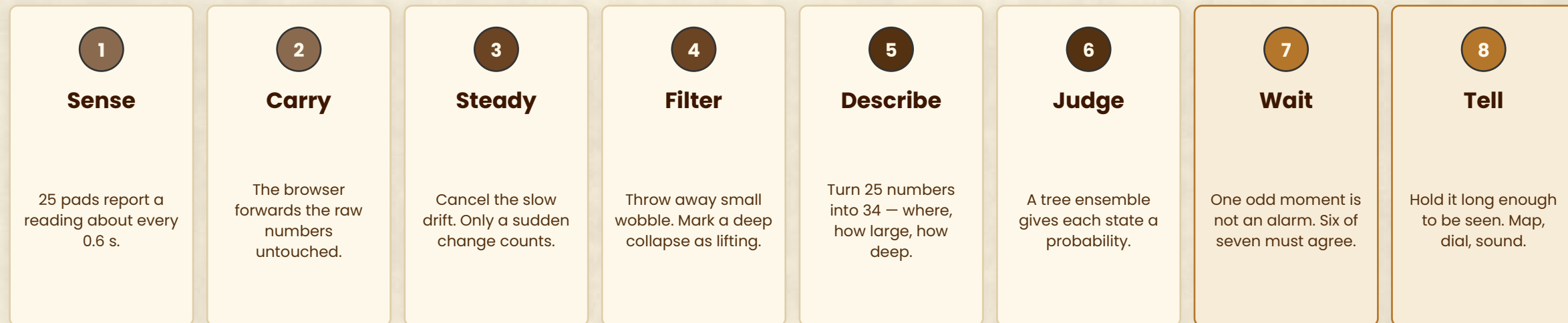
And the separation runs on press magnitude: the heaviest presses produce no pull frames at all. Backwards, for a safety device.

In sample, 13.4% of normal frames already read as pull. The annunciator keeps them off the bedside.

Proposed Method (2/7)

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End-to-end pipeline: where each decision is made



ON THE PATIENT'S SIDE

ON THE SERVER

AT THE BEDSIDE

The browser never decides anything

It moves bytes and draws pixels. Every judgement happens in one place, so the audit trail is complete by construction.

The model is sealed

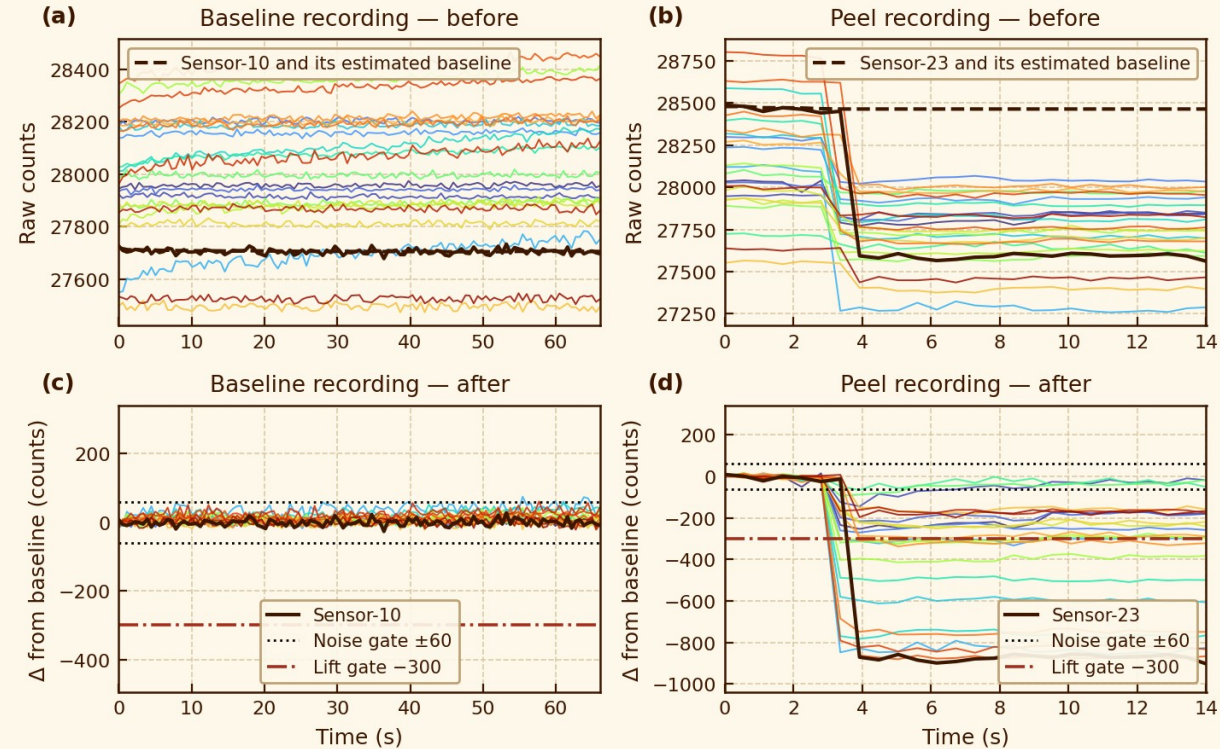
Stored with a fingerprint, so a swapped model is refused rather than quietly used.

Speed is not the constraint

A judgement costs 0.127 ms. The wait for agreement sets the response time — a median of 5.6 s.

Proposed Method (3/7)

Signal conditioning: cancelling drift without swallowing the event



Quiet: stays inside ± 60 for 67 seconds. Peel: crosses -300 within two frames.

The problem

Sweat and body heat move every pad over a shift. A reference taken at start-up is wrong an hour later.

The fix

Each pad carries its own estimate of quiet. Everything downstream works on the distance from it.

The trap it avoids

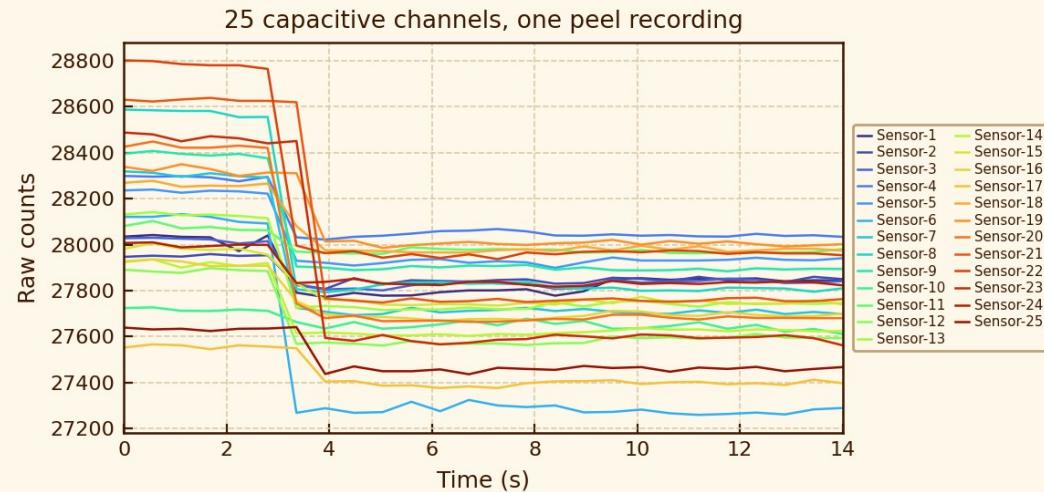
If it chased the reading, a long press would become the new normal. Past 120 counts it stops following and coasts — the flat dashed lines above.

Proposed Method (4/7)

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Feature extraction: the map itself, and nine summaries of it

What the raw signal looks like



Why not just the summaries?

An earlier version kept the nine summaries and threw the map away. At the same alarm budget it catches 70% of pull episodes against 100%.



25 The map itself

One value per pad, left where it is. No summary recovers geometry once averaged away.

4 How big

Smallest, largest, average and spread across the patch.

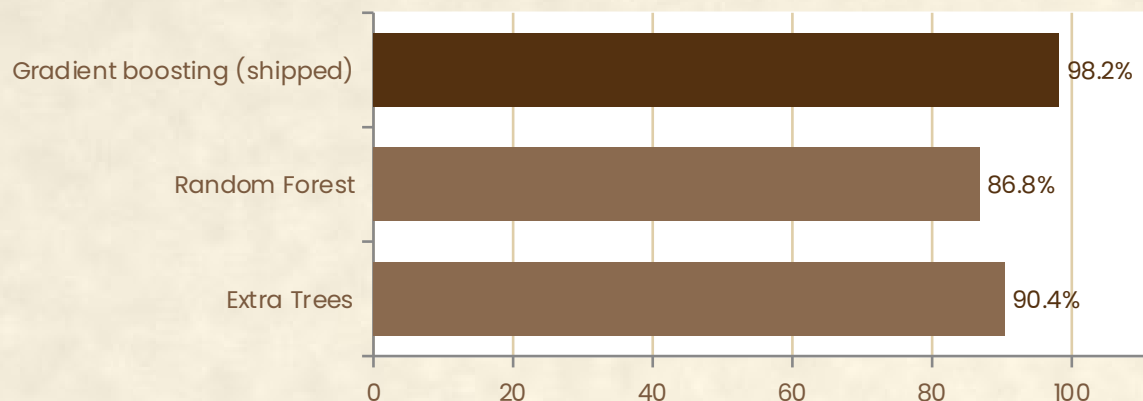
5 How many, and how deep

Pads past five marks — -300, -600, -1000 lifting; +300, +1000 pressing.

Not in the shipped vector: the optional gradient mode is off everywhere. 34 = 25 pads + these nine.

Proposed Method (5/7)

Classifier selection: three candidates, one hundred repeats



Episode sensitivity over 100 grouped-CV repeats — 1,500 fits.

A retraction worth showing

An earlier comparison reported 100.00% accuracy, and that justified shipping it. The feature matrix had every pad duplicated. On the correct 34 columns it scores 97.53%.

A perfect score across 81 files is a bug report. Every figure here carries an interval.

It wins on the metric the device is judged by

98.2% against 86.8% for the forest — and it leads in 100 of 100 repeats.

It is the steadiest

Shortest box on every measure. Moving least when the data changes is worth as much as the best average.

It is fastest where speed is spent

0.127 ms a frame against about 1.04 ms. Slower to train, which a device pays once.

The column it does not lead

The forest is quieter — 6.92 against 7.93 an hour — but pays 11.4 points of sensitivity for it.

Proposed Method (6/7)

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Temporal annunciator: deciding when to speak

Two rules, in plain terms

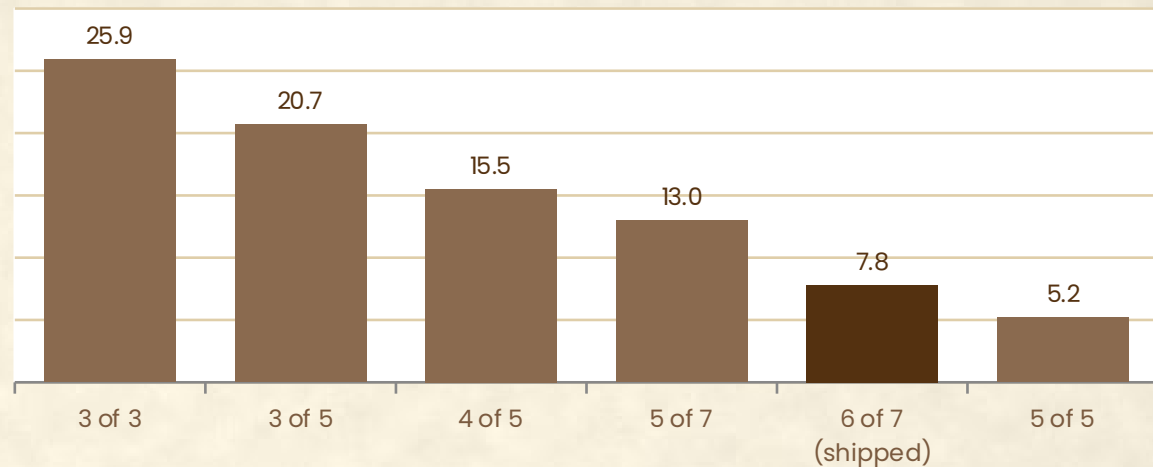
Wait for agreement

Six of the last seven must agree — about 3.4 seconds of evidence.

Then hold it

Held about five seconds, so it cannot be missed or flicker.

Each escalation raised the alarm exactly once.



Every setting holds 100% sensitivity except the last, which drops to 97.5%.

7.8 / hr

Alarm burden at the shipped setting

Down from 13.0 at the previous setting

4.9%

False alarms per recording

95% CI 1.3 – 16.1

5.60 s

Median time from pull to alarm

Worst case 17.36 s

100%

Episode sensitivity held

95% CI 91.2 – 100.0

Is 7.8 alarms an hour acceptable? We cannot claim it is, and we do not.

An intensive-care cohort reports about 5 alarms an hour, and no threshold defining a high rate. So this is roughly double an existing burden — not a figure under a published safe limit.

Proposed Method (7/7)

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Operator console: what a nurse sees

SCREENSHOT TO BE INSERTED BEFORE 7 SEPTEMBER

1600 × 900, light theme, English, replayed to a level-3 frame.

Zero install

Standard browser, hospital tablet or workstation.

No classification in the interface

It renders what the server concluded, and never decides.

Three languages

Thai, English and Japanese, switchable mid-alarm.

1 Capacitive map

Blue where the dressing lifts, red where it is loaded.

2 Cumulative risk dial

A running index, so a slow pull shows before any single frame does.

3 Channel strip chart

All 25 channels, for tracing an alarm back to its pads.

4 Severity banner

Four states, with colour and a frequency-scaled tone.

5 Ward slots

Empty until an operator assigns a bed. No fabricated identity.

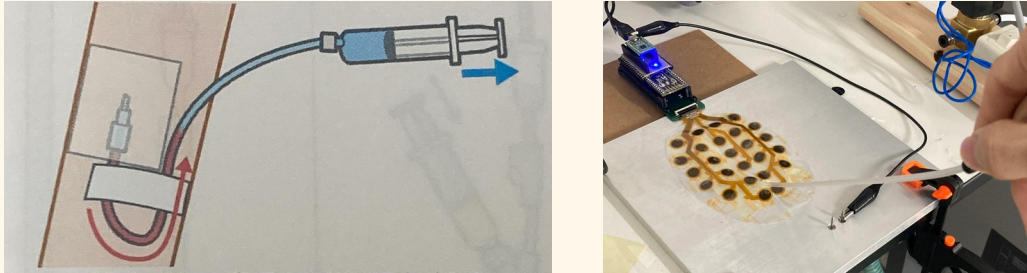
Research prototype. Not assessed against IEC 62304. Not a certified medical device.

Experimental Method

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Bench, fixation and synchronised capture

Fixation and pull mechanics



Slack-loop fixation, drawn at a controlled rate so the mechanics repeat.

Adhesion upgrade — the reason for round 2

- Round 1: vacuum rig, no adhesive, one session, one mounting.
- High-suction source and a reworked backing membrane.
- So shear reaches the pads instead of a loose mount.
- The honest first fix for the horizontal blind spot.

Synchronised capture



Action and response on one timeline, so a disputed frame can be traced back.

TO BE CAPTURED BEFORE 7 SEPTEMBER

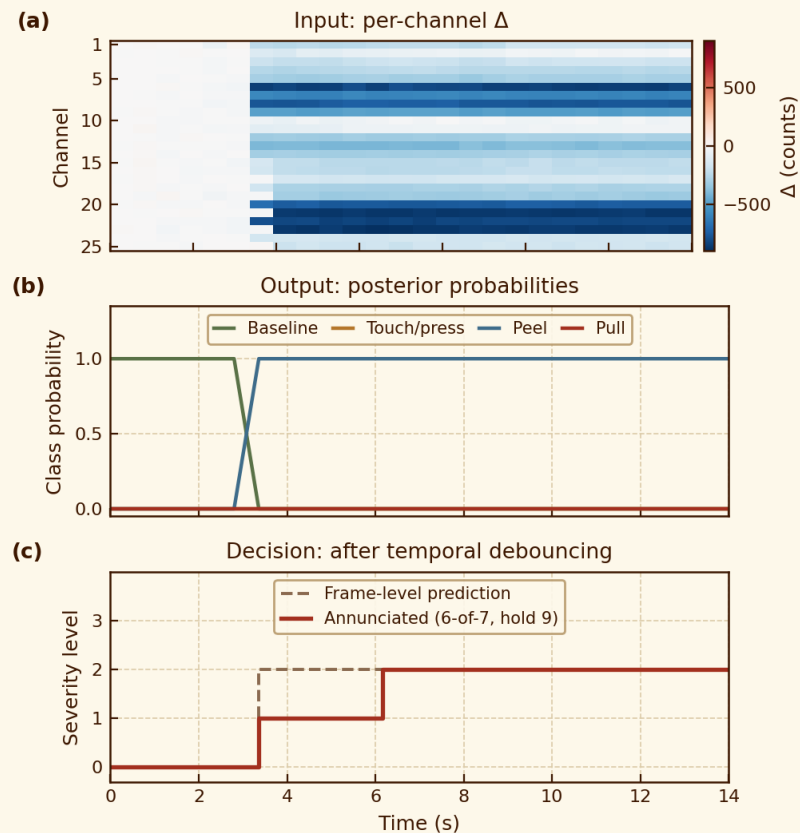
Dual-feed demonstration video

Webcam above, live console below, one clock across both. Two takes: a touch that stays silent, a pull that escalates.

Results (1/4)

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One peel recording, frame by frame through the shipped model



1 Going in — 25 numbers a frame

One change value per pad, per frame. Frames under 60 counts never reach the model.

2 Coming out — four probabilities

A probability for each state, every frame. No serving path hand-codes a level.

3 The decision — agreement, not reaction

The model calls peel at once. The annunciator waits for six of seven — about three seconds.

Results (2/4)

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Quantitative validation over 81 bench recordings

100.0%

Episode sensitivity

95% CI 91.2 – 100.0 · 0 missed of 30 pull files

7.8 / hr

Alarm burden

4.9% of recordings false-alarm · CI 1.3 – 16.1

5.60 s

Median time to alarm

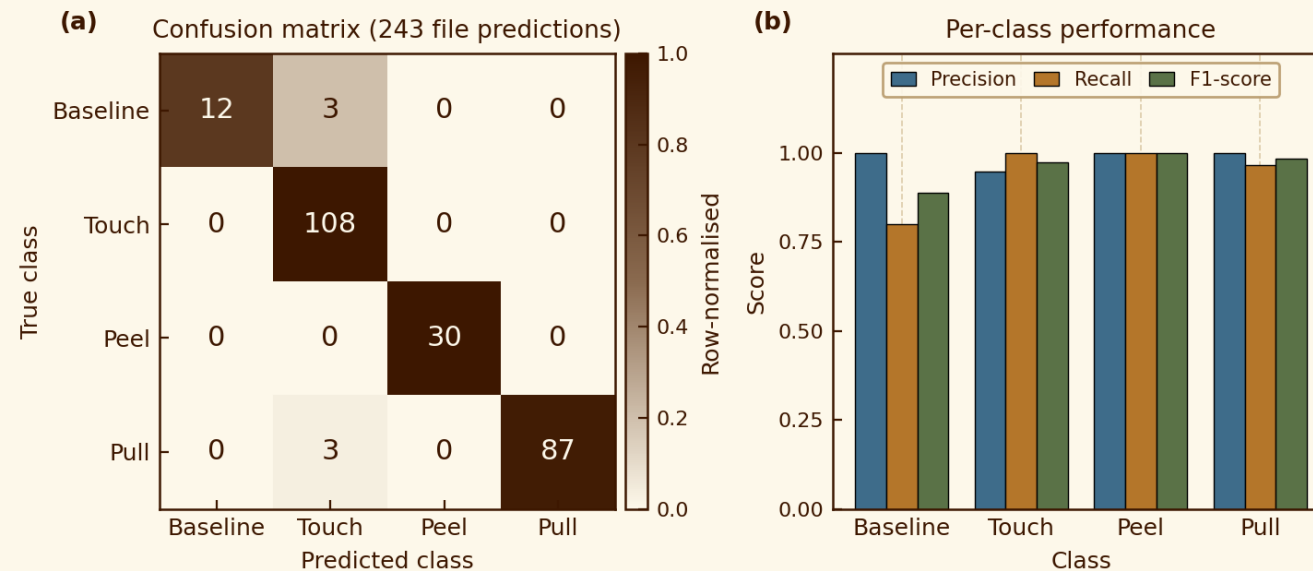
Worst observed 17.36 s · 1.00 onsets / event

97.53%

File-level accuracy

Wilson 95% CI 91.4 – 99.3 · see caveat

Where the errors are, and what each class costs



How to read these two units

Episode level is the clinical unit.

When a pull happened, did the system raise an alarm?
Independent of clip length.

File level is not, and we do not lead with it.

A majority vote across frames, so it moves with clip length. Re-voted at round-2 length, pull recall goes to zero while episode detection stays at one.

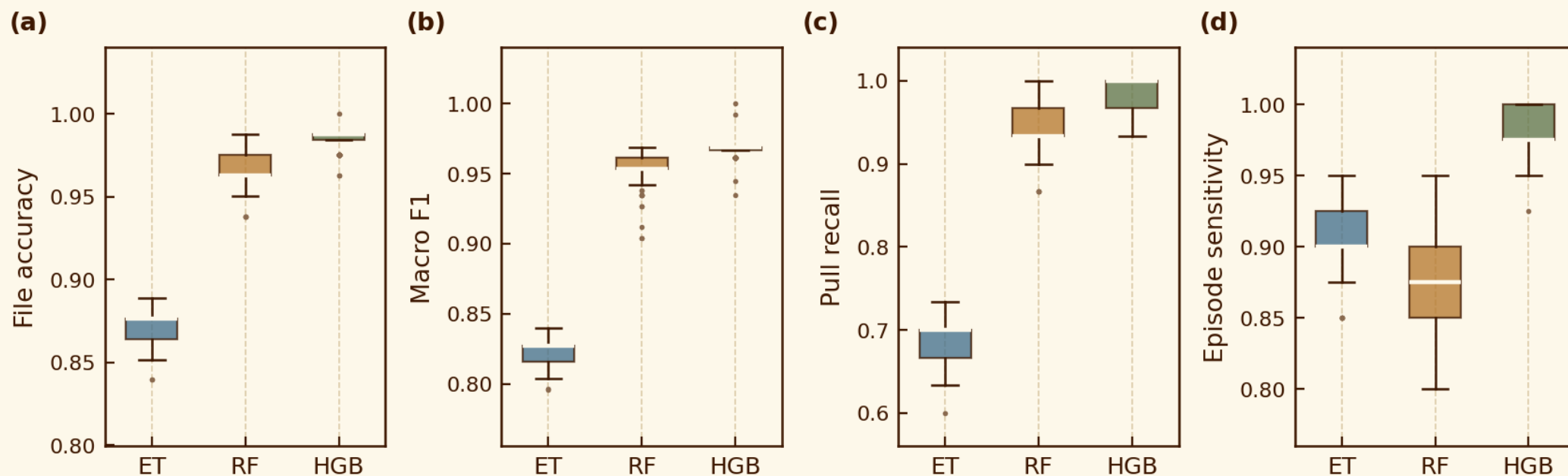
Six errors in 243 predictions. Peel is called perfectly – which on 30 files is not evidence of perfection.

Results (3/4)

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Stability across 100 re-splits

Grouped 5-fold cross-validation, 100 repeats (ET = Extra Trees, RF = Random Forest, HGB = HistGradientBoosting, shipped)



Why re-split at all

With 81 recordings, the split moves the answer more than the estimator does. So it is drawn a hundred times.

What the boxes say

Gradient boosting leads on all four, with the shortest box on every one.

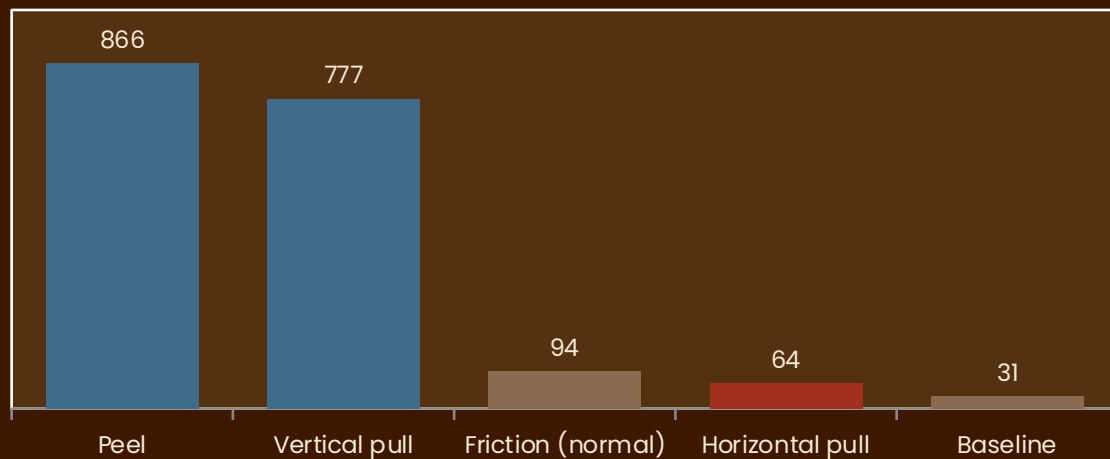
What they do not say

Every dot is one mounting, one session. Stable under re-splitting is not the same as general.

Results (4/4)

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What this does not yet show



How far the reading falls when the dressing leaves the skin. The threshold is 300 counts.

This is continuity, not a new failure

Our own earlier autoencoder reported F1 0.882 vertical against 0.833 horizontal, with false detections.

Two independent methods, same hardware, same wall. The wall is the sensor.

1 Horizontal detachment is invisible to this patch

No lift signal at all. Whatever is flagged there is flagged as a press.

2 One session, one mounting

All 81 recordings share one attachment. Nothing shows it survives re-application.

3 The alarm setting is tuned in-sample

Chosen on this corpus. It must be measured on round 2 without retuning.

4 Spatial output from live hardware is unverified

Confirmed for the recorded corpus, never through the live path.

Discussion

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From relative delta to absolute capacitance

TODAY

Relative reading

Every pad is measured against a baseline taken at start-up — never an absolute value.

- Attached at power-on and detached at power-on give two references for one state.
- So it knows something changed, but not what state it began in.
- A patch already lifted at power-on reads as normal all session.

Resting attached $\approx$ 28,000 counts. Deepest anomaly measured: 27,251.

NEXT GENERATION

Absolute reading

A capacitance-to-digital converter reports each pad in real units — no assumed starting point.

- No start-up calibration, and no dependence on how the patch was sitting.
- Tube placement confirmable at the moment of attachment.
- An already-lifted patch is visible immediately.
- Features and classifier unchanged — only the input units move.

The classifier, the features and the annunciator all survive the change — but every figure in this deck must then be re-measured.

Conclusion

Where this stands, and what has to be measured next

Before 7 September

Dual-feed demonstration video

Webcam and console on one clock.

Not started

This slide deck

English, every figure traceable to the metrics file.

Draft ready

Console screenshot

Light theme, English, level-3 frame.

Not started

Round-2 recordings

First acquisition on the high-suction bench.

In progress

Engineering road ahead

Now

Re-measure the alarm setting on round 2 — without retuning it.

Q4 2026

Port to the absolute readout and re-validate every figure.

Q1 2027

Several mountings, several people, per-session results.

Beyond

Simulated ICU pilot, multi-bed from one station.

Where this stands

Every pull episode caught, at 7.8 alarms an hour, in a median of 5.6 seconds — and it cannot see a horizontal detachment at all, on one mounting, with an interval running down to 91%.

The work ahead is not a better classifier. It is a sensor that responds when the patch slides.